

Times asked: **6 times**

5 times

4 times

3 times

2 times

1 time

indicates 5-mark question

Distributed Computing Question bank

1. Introduction to Distributed Systems

1. Explain the different issues (or challenges) faced in distributed systems. #
2. Explain the goals of distributed systems. #
3. Differentiate between NOS, DOS and Middleware in the design of distributed systems. #
4. What is middleware? Explain the services offered by middleware in distributed systems. #

2. Communication

5. What is Remote Procedure Call? Describe the working of RPC in detail.
6. Explain the various ordered semantics used for Many-to-Many communication with examples.
7. Explain group communication and its types.

3. Synchronization

8. Explain Raymond's tree-based algorithm for distributed mutual exclusion.
9. Explain the Bully election algorithm with a suitable example.
10. What is logical clock? Why are logical clocks required in distributed systems? How Lamport does synchronizes logical clock? Which events are said to be concurrent in Lamport's timestamp.
11. What are physical clocks? Explain any one physical clock synchronization algorithm.
12. What is distributed mutual exclusion? Explain how Suzuki-Kasami's broadcast algorithm achieves distributed mutual exclusion.
13. Explain Ricart-Agrawala's algorithm and how it optimizes message overhead in achieving mutual exclusion.
14. Explain Chandy-Misra-Haas Algorithm for distributed deadlock detection.

4. Resource and Process Management

15. Explain the key features of a Global Scheduling algorithm. #
16. Explain load estimation policies, process transfer policies, and location policies used in load balancing approaches in distributed systems.
17. Explain code migration and its need in distributed systems. Explain the role of process-to-resource binding and resource-to-machine binding in code migration.
18. Explain how load balancing benefits a distributed system.

5. Replication, Consistency and Fault Tolerance

19. What is fault tolerance? Describe different types of failure models.
20. Explain any five data-centric consistency models with example data stores.

6. Distributed File Systems

21. What are the desirable features of a good Distributed File System (DFS)?
22. Explain synchronization in Distributed File Systems and its challenges. #
23. Discuss the Google File System (GFS) as a scalable distributed file system.
24. Explain different file caching schemes used in Distributed File Systems.

Module-wise Marks Weightage and Question Count

	1	2	3	4	5	6
2025 Aug	15 (2)	10 (1)	40 (4)	25 (3)	5 (1)	25 (3)
2025 May	15 (2)	15 (2)	35 (4)	25 (3)	20 (2)	10 (1)
2024 Dec	15 (2)	15 (2)	30 (3)	25 (3)	20 (2)	15 (2)
2024 May	10 (1)	10 (1)	30 (3)	20 (2)	30 (3)	20 (2)
2023 Dec	15 (2)	15 (2)	30 (3)	20 (2)	30 (4)	10 (1)
2023 May	10 (2)	30 (3)	40 (5)	25 (3)	10 (1)	10 (1)
Estimate	15 (2)	15 (2)	30-40 (3-4)	25 (3)	20 (2)	10-15 (1-2)
Total	80	95	205	140	115	90

Asked once:

1. Introduction to Distributed Systems

2. Communication

1. Differentiate between RMI and RPC. #
2. Explain various forms of message-oriented communication with suitable example.

3. Synchronization

3. Explain the need of election algorithm. #
4. Compare two election algorithms and recommend the most efficient one for large-scale distributed systems.

4. Resource and Process Management

5. Explain code migration and its techniques.
6. Describe code migration issues in detail.

5. Replication, Consistency and Fault Tolerance

7. Explain how replication helps in achieving fault tolerance. #
8. Explain how Monotonic read consistency model is different than Read your Write consistency model. #
9. Discuss the technique to achieve Process resilience.
10. Discuss design and implementation issues of distributed shared memory.
11. Write a short note on Replication and the types of it.
12. Discuss and differentiate various client consistency models.

6. Distributed File Systems

13. How are modifications propagated in file caching schemes? #
14. Explain the working of Distributed File System with its applications.

PYQs by Module:

1. Introduction to Distributed Systems

2025 Aug

List and describe five challenges faced in distributed systems. #

Write a short note on Middleware in Distributed System.

2025 May

Enlist the issues in designing Distributed Systems. Explain failure transparency & location transparency in detail. #

Explain the goals of distributed systems.

2024 Dec

What are the services offered by middleware? #

Discuss design issues in distributed systems.

2024 May

Discuss various issues and goals related to design of distributed systems.

2023 Dec

Differentiate between NOS, DOS and Middleware in the design of distributed systems. #

What are different issues and goals related to distributed systems?

2023 May

Explain issues in designing Distributed system. #

Compare NOS and DOS. #

Yellow = Issues in designing distributed systems. (6)

Cyan = Goals of distributed systems. (3)

Green = Middleware and its services. (2)

Grey = Compare NOS, DOS and Middleware. (2)

2. Communication

2025 Aug

Explain RPC in distributed systems.

2025 May

Explain the various ordered semantics used for Many-to-Many communication. #

What is RPC? Explain working of RPC in detail.

2024 Dec

Differentiate between RMI and RPC. #

Explain group communication.

2024 May

What is Remote Procedure Call? Describe the working of RPC in detail.

2023 Dec

What is 1:M and M:1 group communication? #

What is RPC? Explain working of RPC in detail.

2023 May

What is Remote procedure call? Explain how transparency is achieved in RPC.

Explain various forms of message-oriented communication with suitable example.

Explain Absolute ordering and Causal ordering process with the help of example for many to many communication.

Yellow = Remote Procedure Call (RPC): Explain, Working, transparency achieved. (5)

Cyan = Ordering processes in Many-to-Many communication. (2)

Green = Group communication and its types. (2)

3. Synchronization

2025 Aug

Compare two election algorithms and recommend the most efficient one for large-scale distributed systems.

Explain the concept and necessity of logical clocks in distributed systems, and describe Lamport's algorithm.

Describe how Raymond's algorithm ensures mutual exclusion.

Explain the Suzuki-Kasami Broadcast Algorithm for mutual exclusion.

2025 May

Explain Bully Algorithm with example. #

Explain Chandy-Misra-Haas Algorithm.

Explain Raymond's Algorithm for mutual exclusion.

How does Lamport synchronize logical clock, explain with example. Which events are said to be concurrent in Lamport's timestamp.

2024 Dec

Explain Raymond's tree-based algorithm for mutual exclusion.

Explain Ricart-Agrawala's algorithm and how it optimizes message overhead in achieving mutual exclusion.

Explain bully election algorithm.

2024 May

Explain Bully election algorithm.

What is distributed mutual exclusion? Explain how Suzuki-Kasami's broadcast algorithm achieves distributed mutual exclusion.

What is physical clock? Explain any one physical clock synchronization method.

2023 Dec

Explain Bully election algorithm with suitable example.

Explain Raymond's algorithm for mutual exclusion.

What are physical clocks? Explain any one physical clock synchronization algorithm.

2023 May

Explain the need of election algorithm. #

Justify how Ricart-Agrawala's algorithm optimized the Message overhead in achieving mutual exclusion. #

What is logical clock? Why are logical clocks required in distributed systems? How Lamport does synchronizes logical clock? Which events are said to be concurrent in Lamport's timestamp.

Explain Chandy-Misra-Hass Algorithm for distributed deadlock detection.

Discuss Raymond's tree-based algorithm of token based in distributed mutual exclusion.

Yellow = Raymond's Algorithm for mutual exclusion. (5)

Cyan = Election Algorithm: Bully algorithm (4)

Green = Logical Clocks (Lamport Clock, Need, Concurrency) (3)

Grey = Physical Clocks, explain any synchronization algorithm. (2)

Red = Suzuki-Kasami Broadcast Algorithm for mutual exclusion. (2)

Blue = Chandy-Misra-Haas Algorithm for Deadlock detection. (2)

Pink = Ricart-Agrawala Algorithm for optimizing message overhead for mutual exclusion. (2)

4. Resource and Process Management

2025 Aug

Elaborate key Features of Global Scheduling algorithm. #

Explain how load balancing benefits a distributed system.

Write a short note on Code Migration in Distributed System.

2025 May

What are the key features of Global Scheduling algorithm. #

Explain load estimation, process transfer and location policies with respect to load balancing approach in distributed systems.

What is need of code migration? Explain the role of process to resource binding and resource to machine binding in code migration.

2024 Dec

Explain key features of Global Scheduling algorithm. #

Justify how load balancing is useful in distributed systems.

Discuss code migration in distributed system.

2024 May

Explain different load estimation policies & process transfer policies used in load balancing.

What is need of code migration? Explain the role of process to resource binding and resource to machine binding in code migration.

2023 Dec

Explain code migration and its techniques.

Explain different load estimation & process transfer policies used by load balancing algorithms.

2023 May

Explain desirable features of global scheduling algorithm. #

Explain different load estimation & process transfer policies used by load balancing algorithms..

Describe code migration issues in detail.

Yellow = Features of Global Scheduling algorithm. (4)

Cyan = Load Balancing: Load Estimation, Process Transfer and (once) Location policies. (4)

Green = Code migration: Need, Role of process-resource-machine binding. (4) // keeping techniques and issues separate.

Grey = Benefits of load balancing. (2)

5. Replication, Consistency and Fault Tolerance

2025 Aug

Explain how replication helps in achieving fault tolerance. #

2025 May

What is fault tolerance? Describe different types of failure models.

Discuss the technique to achieve Process resilience.

2024 Dec

What is fault tolerance? Explain failure models.

Explain any five data-centric consistency models.

2024 May

Explain any five data-centric consistency models with example data stores.

Discuss design and implementation issues of distributed shared memory.

What is fault tolerance? Describe different types of failure models.

2023 Dec

Explain how Monotonic read consistency model is different than Read your Write consistency model. #

What is fault tolerance? Explain failure models. #

What are different data centric consistency models?

Write a short note on Replication and the types of it.

2023 May

Discuss and differentiate various client consistency models.

Yellow = Fault tolerance, types of failure of models. (4)

Cyan = Five data-centric consistency models. (3)

6. Distributed File Systems

2025 Aug

Explain synchronization in DFS with its challenges. #

Analyze the scalability of the Google File System (GFS).

Explain the working of Distributed File System with its applications.

2025 May

Desirable features of a good DFS?

2024 Dec

Explain synchronization in DFS with its challenges. #

Discuss Google File System (GFS) as a scalable distributed file system.

2024 May

What are desirable features of a good DFS?

Explain various file caching schemes.

2023 Dec

Explain file caching schemes.

2023 May

List desirable features of distributed File system. How are modifications propagated in file caching schemes?

Yellow = Desirable Features of a Good DFS. (3)

Cyan = Scalability of Google File System (GFS). (2)

Green = File caching schemes (2)

Grey = Synchronization in DFS and its challenges. (2)

Syllabus

Module	Content	Hrs
1	Introduction to Distributed Systems	4
1.1	Characterization of Distributed Systems: Issues, Goals, Types of distributed systems, Grid and Cluster computing Models, Hardware and Software Concepts: NOS, DOS.	
1.2	Middleware: Models of middleware, Services offered by middleware.	
2	Communication	4
2.1	Interprocess communication (IPC): Remote Procedure Call (RPC), Remote Method Invocation (RMI).	
2.2	Message-Oriented Communication, Stream Oriented Communication, Group Communication.	
3	Synchronization	10
3.1	Clock Synchronization: Physical clock, Logical Clocks, Election Algorithms	
3.2	Distributed Mutual Exclusion, Requirements of Mutual Exclusion Algorithms and Performance measures. Non- token Based Algorithms: Lamport, Ricart–Agrawala’s and Maekawa’s Algorithms; Token-based Algorithms: Suzuki-Kasami’s Broadcast Algorithms and Raymond’s Tree-based Algorithm; and Comparative Performance Analysis.	
3.3	Deadlock: Introduction, Deadlock Detection: Centralized approach, Chandy - Misra_Hass Algorithm.	
4	Resource and Process Management	7
4.1	Desirable Features of Global Scheduling algorithm, Task assignment approach, Load balancing approach and load sharing approach.	
4.2	Introduction to Process Management, Process Migration, Code Migration.	
5	Replication, Consistency and Fault Tolerance	
5.1	Distributed Shared Memory: Architecture, design issues.	8
5.2	Introduction to replication and consistency, Data-Centric and Client-Centric Consistency Models, Replica Management.	
5.3	Fault Tolerance: Introduction, Process resilience, Recovery.	
6	Distributed File Systems	6
6.1	Introduction and features of DFS, File models, File Accessing models, File-Caching Schemes, File Replication, Case Study: Network File System (NFS).	
6.2	Designing Distributed Systems: Google Case Study.	

PYQs (Marked by module)

Module 1: Introduction to Distributed Systems

Module 2: Communication

Module 3: Synchronization

Module 4: Resource and Process Management

Module 5: Replication, Consistency and Fault Tolerance

Module 6: Distributed File Systems

2025 Aug

Q 1

-  A Explain how replication helps in achieving fault tolerance. [5]
-  B List and describe five challenges faced in distributed systems. [5]
-  C Elaborate key Features of Global Scheduling algorithm. [5]
-  D Explain synchronization in DFS with its challenges. [5]

Q 2

-  A Explain RPC in distributed systems. [10]
-  B Analyze the scalability of the Google File System (GFS). [10]

Q 3

-  A Compare two election algorithms and recommend the most efficient one for large-scale distributed systems. [10]
-  B Explain the concept and necessity of logical clocks in distributed systems, and describe Lamport's algorithm. [10]

Q 4

-  A Describe how Raymond's algorithm ensures mutual exclusion. [10]
-  B Explain the working of Distributed File System with its applications. [10]

Q 5

-  A Explain how load balancing benefits a distributed system. [10]
-  B Explain the Suzuki-Kasami Broadcast Algorithm for mutual exclusion. [10]

Q 6 Write a short note on

-  A Code Migration in Distributed System [10]
-  B Middleware in Distributed System [10]

Q 1

- A** Enlist the issue in designing the Distributed System? Explain failure transparency & location transparency in detail [5]
- B** What are the key features of Global Scheduling algorithm. [5]
- C** Explain Bully algorithm with example [5]
- D** Explain the various ordered semantics used for Many to Many communication [5]

Q 2

- A** Explain Chandy Misra Hass Algorithm [10]
- B** Desirable features of a good DFS? [10]

Q 3

- A** Explain load estimation, process transfer and location policies with respect to load balancing approach in distributed systems. [10]
- B** Explain Raymond's algorithm for mutual exclusion. [10]

Q 4

- A** Explain the goals of distributed systems. [10]
- B** How Lamport does synchronizes logical clock explain with example? Which events are said to be concurrent in Lamport's timestamp. [10]

Q 5

- A** What is fault tolerance? Describe different types of failure models. [10]
- B** What is RPC? Explain working of RPC in detail [10]

Q 6

- A** Discuss the technique to achieve the Process resilience. [10]
- B** What is need of code migration? Explain the role of process to resource and resource to machine binding in code migration. [10]

Q 1

- A Explain key Features of Global Scheduling algorithm. [05]
- B What are the services offered by middleware? [05]
- C Differentiate between RMI and RPC. [05]
- D Explain synchronization in DFS with its challenges. [05]

Q 2

- A Justify how load balancing is useful in distributed system. [10]
- B What is fault tolerance? Explain failure models. [10]

Q 3

- A Explain group communication. [10]
- B Explain Raymond's tree based algorithm for mutual exclusion. [10]

Q 4

- A Explain Ricart-Agrawala's algorithm and how it optimizes the message overhead in achieving mutual exclusion. [10]
- B Discuss Google file system (GFS) as a scalable distributed file system. [10]

Q 5

- A Explain bully election algorithm. [10]
- B Discuss code migration in distributed system. [10]

Q 6

- A Explain any five data centric consistency models. [10]
- B Discuss design issues in distributed systems. [10]

Q1

A Explain any five data centric consistency models with example data stores. [10]

B Explain different load estimation policies and process transfer policies used in load balancing approach of distributed system. [10]

Q2

A What is Remote Procedure Call? Describe the working of RPC in detail. [10]

B Explain Bully election algorithm. [10]

Q3

A Discuss design and implementation issues of distributed shared memory. [10]

B What are desirable features of a good DFS? [10]

Q4

A Discuss various issues and goals related to design of distributed system. [10]

B What is distributed mutual exclusion? Explain how Suzuki-Kasami's broadcast algorithm achieves distributed mutual exclusion. [10]

Q5

A What is need of code migration? Explain the role of process to resource and resource to machine binding in code migration. [10]

B Explain various file caching schemes. [10]

Q6

A What is physical clock? Explain any one physical clock synchronization method. [10]

B What is fault tolerance? Describe different types of failure models. [10]

Q 1

- A Explain how Monotonic read consistency model is different than Read your Write consistency model [5]
- B What is 1:M and M:1 group communication? [5]
- C Differentiate between NOS DOS and Middleware in the design of distributed systems [5]
- D What is fault tolerance? Explain failure models. [5]

Q 2

- A Explain code migration and its techniques. [10]
- B Explain Bully election algorithm with suitable example. [10]

Q 3

- A Explain Raymond's algorithm for mutual exclusion. [10]
- B What are different data centric consistency models [10]

Q 4

- A Explain different load estimation and process transfer policies used by load balancing algorithms. [10]
- B What are physical clocks? Explain any one physical clock synchronization algorithm. [10]

Q 5

- A What are different issues and goals related to distributed systems? [10]
- B Explain file caching schemes [10]

Q 6

- A Write a short note on Replication and the types of it. [10]
- B What is RPC? Explain working of RPC in detail [10]

- 1 Attempt any FOUR [20]
 - a Explain issues in designing Distributed system
 - b Compare NOS and DOS
 - c Explain desirable features of global scheduling algorithm
 - d Explain the need of election algorithm.
 - e Justify how Ricart-Agrawala's algorithm optimized the Message overhead in achieving mutual exclusion
- 2 a What is Remote procedure call? Explain how transparency is achieved in RPC [10]
 - b Explain various forms of message oriented communication with suitable example [10]
- 3 a What is logical clock? Why are logical clocks required in distributed systems? How Lamport does synchronizes logical clock? Which events are said to be concurrent in Lamports timestamp [10]
 - b Explain Chandy -Misra_Hass Algorithm for distributed deadlock detection. [10]
- 4 a Explain different load estimation and process transfer policies used by load balancing algorithms. [10]
 - b Describe code migration issues in details [10]
- 5 a Discuss and differentiate various client consistency models. [10]
 - b Explain Absolute ordering and Casual ordering process with the help of example for many to many communication. [10]
- 6 a List desirable features of distributed File system. How are modifications propagated in file caching schemes? [10]
 - b Discuss Raymonds tree based algorithm of token based in distributed mutual exclusion [10]